Constraints on BH – Host Relation at 5 < z < 6.5 using JWST COSMOS-3D

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Background

The relation between supermassive black hole (SMBH) mass and host-galaxy stellar mass provides key evidence for the co-evolution of black holes and galaxies. While this relation is well established in the local Universe, its form at early cosmic times remains poorly constrained.

At z>5, during the early phase of rapid SMBH growth and host galaxy assembly, these systems are ideal laboratories for testing whether the BH–host galaxy relation already exists in the early Universe and for understanding how this relation is established.

Recently discovered “Little Red Dots” (LRD) are compact red objects mainly found at high-z, which exhibit a “V-shaped” spectral energy distribution (SED) and appear to contain SMBH, possibly providing a direct view of an important growth phase in the assembly of the earliest SMBHs.

Data/Sample

The BLAGN sample is selected from a JWST wide field survey: COSMOS-3D. COSMOS-3D combines multi-band NIRCam and MIRI photometry with NIRCam/WFSS F444W slitless spectroscopy over 0.3deg^2, which enables us to select bright, red, and compact targets with emission lines that we visually inspect to select a pure broad-line Hα sample.

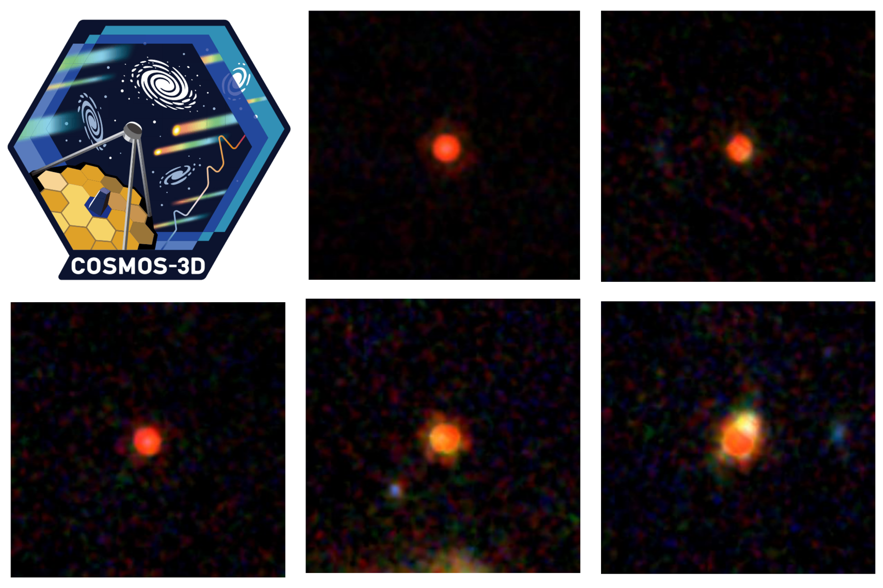

Figure 1. NIRCam RGB example cutouts of the BLAGN sample along with the JWST COSMOS-3D survey logo.

83 BLAGN

53 LRD

Out of 83 BLAGN in our sample, 53 satisfy a standard LRD color selection.

Spectral Analysis

The NIRCam/WFSS F444W spectra is used to fit the Hα broad-line with narrow+broad components and obtain the line properties. The black hole masses are then calculated using single epoch virial BH mass estimation from the Hα broad-line width and luminosity.

Some BLAGN exhibit Balmer absorption, indicative of very dense gas. This is a rare phenomenon at low-z (<0.1% of AGN), but more common at high-z (10-20% of BLAGN). The origin of the absorption is still debated. In our sample, out of the 10 BLAGN that show strong Balmer absorption, 9 fall in the LRD subsample.

90% of Balmer absorption in LRD

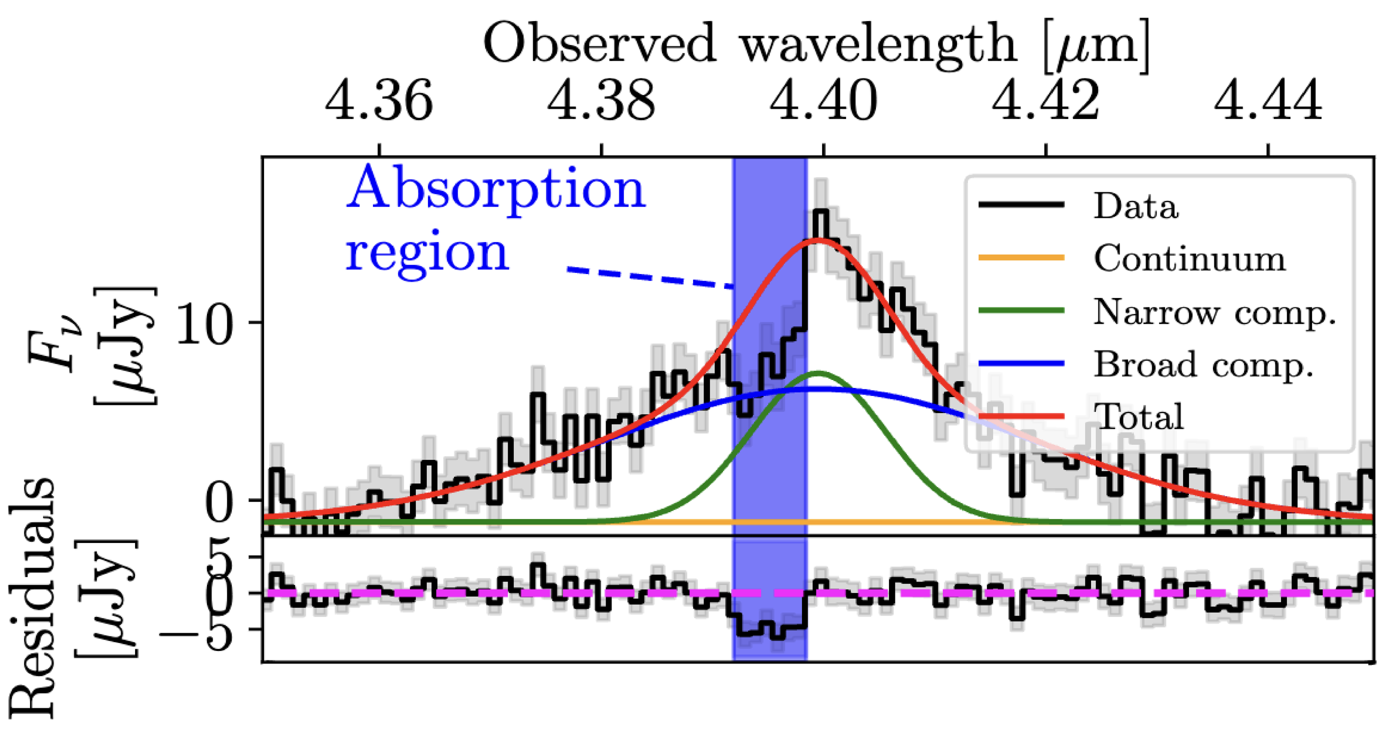

Figure 2. Spectral modelling example of a BLAGN exhibiting Balmer absorption.

Online version

Spatial Decomposition and SED Fitting

To isolate the flux from the host galaxy, we perform a 2D spatial decomposition using GALIGHT. We assume a Sérsic ellipse profile for the host and a simple scaled point-source for the AGN. This fitting is performed in all six NIRCam bands (F115W,F150W,F200W,F277W,F356W,F444W).

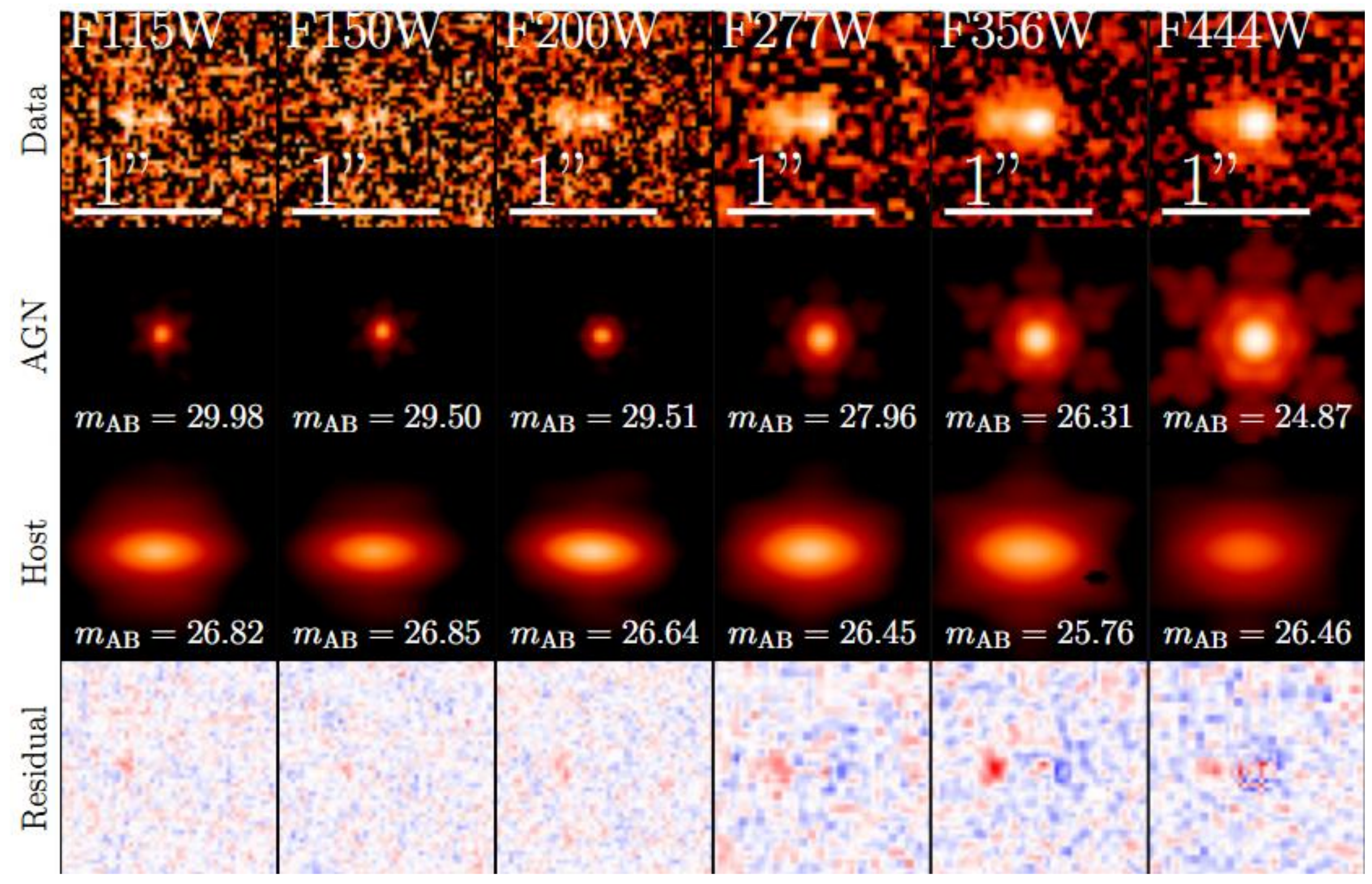

Figure 3. AGN+Host spatial decomposition example for the six NIRCam filters of the same target.

To constrain the stellar mass, the host galaxy photometry from the decomposition is fitted using BAGPIPES with a delayed tau formation history.

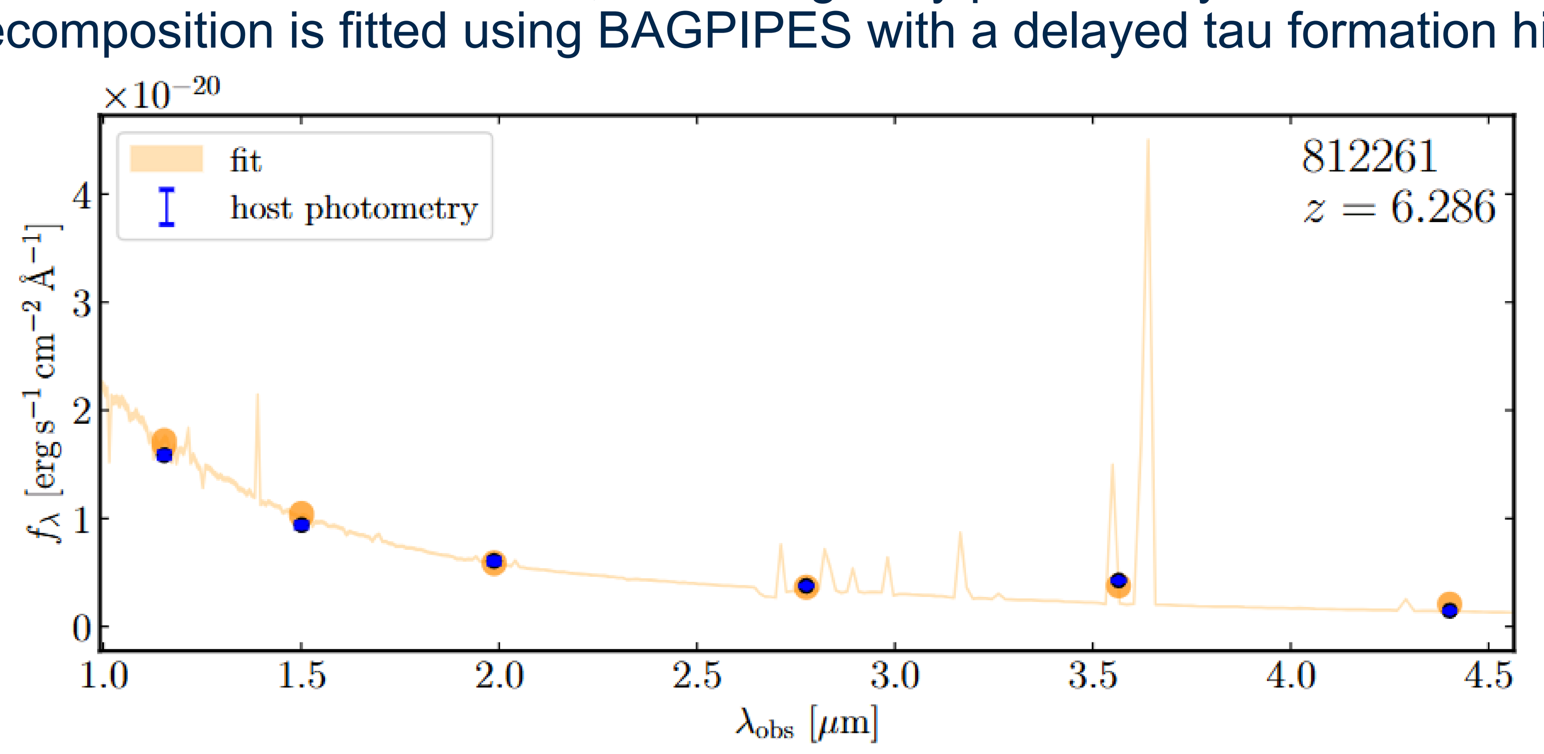

Figure 4. BAGPIPES SED modelling example for a BLAGN.

Results

- Host galaxies contribute significantly to the rest-frame UV flux, but the AGN dominates in the rest-frame optical.
- High-z BLAGN show a large scatter in the $M_{BH} - M_\star$ plot, showing the local relation isn’t formed yet, or has a much larger scatter. The BLAGN also showcase higher $M_{BH}/M_\star$ ratio.
- Both Balmer-absorption BLAGN and LRD have more massive BH than regular BLAGN and there is a large overlap between the two subsamples.

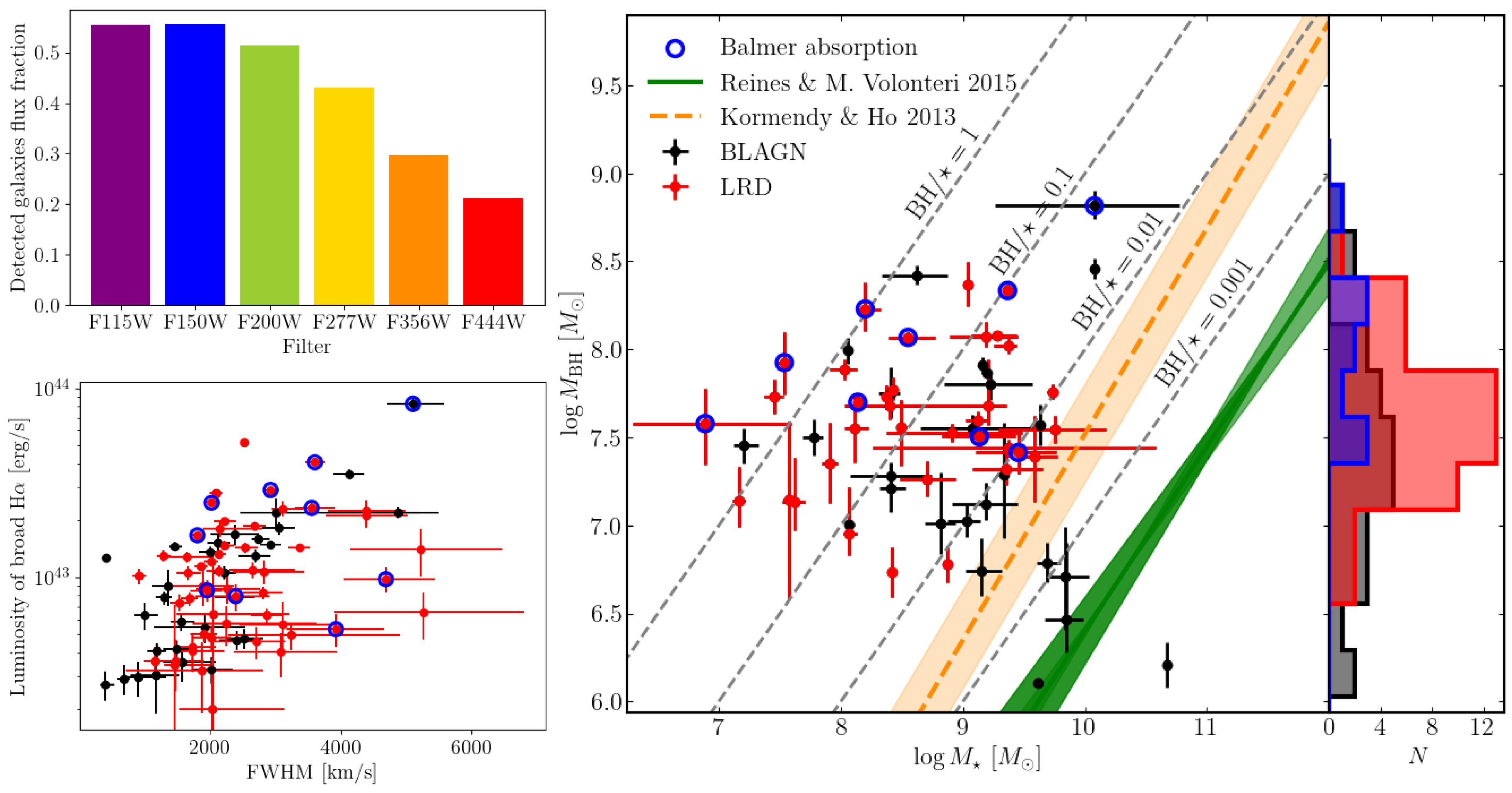

Figure 5. Top left: Host galaxy flux contribution for the different filters, showing a decrease towards redder wavelengths. Bottom left: Spectral properties used to measure the BH mass. Right: Black hole mass – Stellar mass relation for the BLAGN sample, along with the BH mass histogram.